

12. Uncertainty

Introduction

The role of uncertainties related to petrophysical (and other) evaluations is often underestimated. In many cases it is wise to do a sensitivity study, i.e. to calculate what the impact is of variations of the input parameters (within their error bars) on the output results.

The sensitivity studies are often made using Monte Carlo methods, i.e. generating many possible models resulting from statistically probable combinations of variations of the input parameters. The output results together build a cumulative distribution function for the major output parameters.

The hydrocarbon volume HCVOL of a reservoir is determined from:

$$\text{HCVOL} = A * h * (\text{N/G}) * \phi * (1 - S_w)$$

where:

A = area of the reservoir

h = thickness of the reservoir (gross)

N/G = net over gross ratio

ϕ = porosity (fraction)

S_w = water saturation (fraction)

$S_{hc} = 1 - S_w$ = hydrocarbon saturation (fraction of pore volume)

The input parameters are average parameters for the prospect or field with an uncertainty distribution around these averages. *Figure 1* shows this graphically (note that the equation shown is the STOIIP equation which is the same as the equation above but corrected to surface condition, using B_o).

Below some general thoughts and considerations are given on the accuracy/uncertainty in each of those parameters. Assessment of uncertainties (and related accuracy/precision) and quality control is an essential task of each petrophysicist (petroleum engineer) and is often valued too low. Money can be made by addressing these issues more seriously.

A: Reservoir Area

The reservoir area "A" is mainly determined from seismic, combined with geological knowledge, e.g. obtained in the region from other wells or outcrop. Well / production tests, combined with reservoir simulation also play an important role in determining the extent of a reservoir. Geological and reservoir engineering modelling tools can play an important role in addition to the seismic interpretation software. Seismic is constrained by a limited resolution and by various assumptions in the modelling (velocity model, migration, etc.). Geological models are based on empirical knowledge and are therefore statistical rather than deterministic. It is probably fair to say that in general the reservoir area is one of the most uncertain factors in the determination of the hydrocarbon volume.

$h^*(N/G)$: Reservoir Thickness and Net-over-Gross

The reservoir thickness h and net-over-gross-ratio (N/G) are normally obtained from logs and cores. Seismic information does not have sufficient vertical resolution to play an important role in this. On the other hand, the extrapolation of reservoir thickness away from the wellbore can only be made on the basis of seismic ("lateral prediction"), because of the limited depth of investigation of logging tools (see further below). Hence, the net reservoir thickness is best known at and around the wellbore and becomes increasingly more uncertain away from the wellbore.

Again, geological modelling has an important role to play here. Additional information can be obtained from capillary pressure curves and wireline formation tests (pressure versus depth graphs). The net-over-gross ratio (N/G) or net sand thickness $h^*(N/G)$ are often obtained from the gamma-ray, using a simple cut-off criterion (e.g. all intervals with GR reading less than x API are sands). In general it is better not to work with cut-offs in this stage, but to only apply them at the final evaluation stage. Also, use of other logs in addition to the gamma-ray, in particular use of the density/neutron combination, often gives a much better discrimination between shales and sands.

An important point to note is that even with wireline logs the net reservoir thickness determination may be inaccurate because of the limited vertical resolution of logging tools. Especially in laminated sand/shale sequences, e.g. turbidites, logs often fail to discriminate reservoir rock from non-reservoir rock (see discussion of "water saturation" below). Currently, this problem can only be solved by use of advanced logging tools (e.g. NMR, FMI) in addition to the standard tools, combined with logging tool response modelling.

Φ : Porosity (related properties: permeability / producibility)

The reservoir porosity ϕ is most often determined from the density log, sometimes combined with neutron or sonic logs. The parameters required in the conversion from density to porosity are best determined from core calibration measurements in the laboratory (taking into account the effect of stress / compaction). An important parameter is the fluid density. Hence, it is important that the fluid type has been determined correctly (gas/oil differentiation) in order to arrive at a good porosity calculation. In order to get a good estimate for the matrix density the lithology has to be known properly.

Note that density tools (especially LWD tools) are rather sensitive for borehole and mudcake conditions, and for lithology (e.g. shale). This sensitivity should be taken into account when an error analysis is carried out on the porosity determination from the density.

Permeability so far has been traditionally determined in the laboratory from core measurements. The main problem here is how to relate these small size plug measurements to the large size grid blocks ("upscaling"). This problem relates to all core (and log) measurements: it is also related to the lateral prediction problem mentioned above.

Another problem inherent to the use of core measurements is the question how representative the core is for the reservoir (core disturbance, biased sampling, laboratory versus reservoir conditions, etc.).

Sw: Water Saturation (related property: fluid type)

The water (and hence hydrocarbon) saturation is mainly determined from resistivity logs, although it can be (and often is) determined independently from capillary pressure curves combined with wireline formation tests.

Resistivity tools are heavily affected by environmental effects, i.e. effects of borehole, mud filtrate invasion, formation layering and borehole deviation / dip. This is so because resistivity tools have been designed to read relatively deep into the formation (a few metres which affects the vertical resolution, i.e. they are affected by layers several meters away. Hence a thin (few feet) oil bearing layer surrounded by low resistivity shoulder beds (e.g. water bearing sands or shales) it is very difficult to "see".

A related problem is the response in laminated sand/shale sequences, e.g. turbidites: the response is dominated by the low resistivity shale laminae, obscuring the hydrocarbon effect in the sand laminae. Therefore, the resistivity response in laminated sequences is often (much) too low.

An additional problem is that saturation can only be determined from resistivity using a resistivity interpretation model, e.g. Archie, Waxman-Smiths. The uncertainties there are: which model to use, and what parameters (m , n , Q_v , etc.) to use in the model. For this purpose laboratory measurements are often required, leading to above mentioned (core measurement related) uncertainties. There are many uncertainties related to the determination of fluid type. The best method is plain testing, but that is often too expensive. Gas/oil determination from logs is often more an art than a science (e.g. the neutron / density separation is often obscured by shale). Again, NMR may have some promise in this area.

Sw: Saturation-Height-Functions

In the HCVOL (or STOIP/GIIP) equation S_w is often expressed as a saturation-height function rather than a single average S_w value (*Figure 2*). This is done for two reasons:

- to ensure a better representation of the transition zone (which can be relative long in e.g. carbonates)
- to ensure consistency between Volumetrics and the Static and Dynamic Models (which also use saturation-height functions).

The saturation-height function(s) are determined from capillary pressure curves or from the log derived saturations.

Cut-Offs

In the calculation of STOIP and GIIP (i.e. the "storage" space) it is advised not to use cut-offs which relate to permeability (i.e. porosity cut-off on the basis of a 1mD cut-off). Usually a distinction is made between sands and shales (or between obvious shales in carbonates sections) on the basis of a V_{shale} cut-off (*Figure 3*) but no subsequent porosity cut-off or S_w cut-off should be made. It is best to handle reservoir vs. non-reservoir through the use of the recovery factor.

Some General Remarks

Some general factors, affecting most of the above determinations are the following ones:

- tool failures / malfunctioning (especially of course if unnoticed).
- depth matching problems.
- statistics, especially with nuclear tools, where count-rate deteriorates with faster logging.
- borehole irregularities / washouts.
- influence of fractures, overpressures, anisotropies / inhomogeneities.

Difference must be made between *systematic and random errors*. Systematic errors are those where the measurement is systematically off (shifted) with respect to the true value, e.g. the measurement is systematically too high (or too low). A random error is an error of which the influence diminishes when the measurement is repeated more often, because the average is close to the truth (refer *Figure 4*).

Because of the many factors and uncertainties involved in the determination of hydrocarbon volume, random errors often cancel each other. Therefore, a systematic error is in most cases by far the most severe one. Such systematic errors can be the result of applying the wrong tool (measurement principle), tool malfunctioning, influence of disturbing factors not properly accounted for, etc.

As mentioned earlier the HCVOL equation expects parameters which are the average for the field or prospect. This means that a straight forward parameter average from the wells is often not sufficient: the geology (trend) has to be included.

Points to Observe

In general the accuracy / precision of the determinations can be improved if the following points are accounted for:

- All tools should be properly calibrated.
- All logs should be properly environmentally corrected if required; especially mud additives (e.g. barytes, KCl) can heavily influence certain measurements.
- Beware of contractor processing: they may have filtered some logs without telling you.
- Beware of possible plain errors (wrong data files, digitizing, wrong well header etc.).
- In thinly bedded reservoirs care has to be taken that the thin bed effect on the resistivity is accounted for. Check with modelling wherever possible.
- Tools and their responses are limited by their measurement principle (physics): they can't measure more than they were designed for (do not believe all contractor claims on what their tools can measure: those claims are only valid for optimal cases).
- In general, more modern tools have more detectors (array tools, imaging tools) and therefore provide more information.
- Tool responses in hostile (HPHT) or non-standard (slim hole, horizontal hole) environments may be less reliable.

Always ask yourself: can I trust this data. Sometimes it is better to have no data (hence, using estimates, literature correlations, etc.) than using wrong data. Make

a habit of checking/questioning: e.g. if the matrix density has been determined to be 2.3 g/cc, ask yourself: can that be right ?? Make a habit of asking for error bars; make audits, compare contractors regularly, check repeat runs. Check log response in known lithologies. Contractors have Quality Control Reference Manuals which they should use.

Example Uncertainty Calculation

Suppose we have a sandstone reservoir with the following properties:

- porosity $\phi = 0.20$
- water saturation $S_w = 0.30$
- thickness $h = 7.17$ metres

This gives an equivalent hydrocarbon column $= \phi \cdot (1 - S_w) \cdot h = 1.00$ metre.

We will get the right evaluation if we use the true formation parameters given below:

- true formation resistivity $R_t = 20.4$ Ohmm
- $n = 1.5$
- $m = 1.5$
- formation water resistivity $R_w = 0.30$ Ohmm
- matrix density $\rho_{ma} = 2.68$ g/cc
- fluid density $\rho_{fl} = 0.74$ g/cc

However, if in our calculations we erroneously take wrong values for these parameters, our evaluation will yield the following (it is a good exercise to check this yourself):

- | | |
|--|--------------------|
| - $R_t = 10$ instead of 20.4 gives | EQHC = 0.74 metres |
| - $n = 2$ instead of 1.5 gives | EQHC = 0.85 metres |
| - $m = 2$ instead of 1.5 gives | EQHC = 0.70 metres |
| - $R_w = 0.15$ instead of 0.30 gives | EQHC = 1.16 metres |
| - matrix density = 2.65 instead of 2.68 gives | EQHC = 0.91 metres |
| - fluid density = 1.0 instead of 0.74 gives | EQHC = 1.23 metres |
| - all of these errors instead of all of the correct values gives | EQHC = 0.68 metres |

Taking a too low R_t yields a too low hydrocarbon saturation hence a too low EQHC. The same goes for a too high value of n . The other variations are equally indicative for the role of those parameters. Clearly, in this example, it is especially important to obtain correct values of R_t (hence the impact of e.g. resistivity modelling / inversion!), m , and fluid density.

In a Monte Carlo study one would do calculations with many combinations of above parameters in the band of uncertainty around their average values. This would give a good impression of the accuracy of the EQHC calculation from a petrophysical point of view.

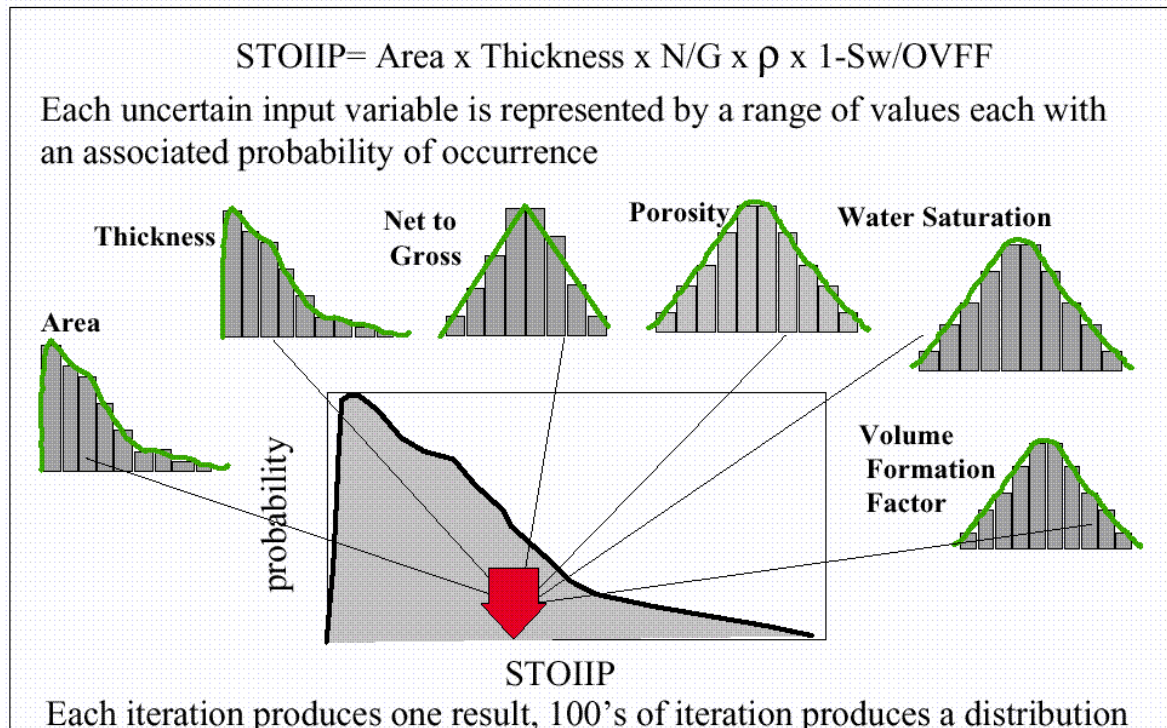


Figure 1: The STOIIP equation (expecting average parameters with an uncertainty distribution)

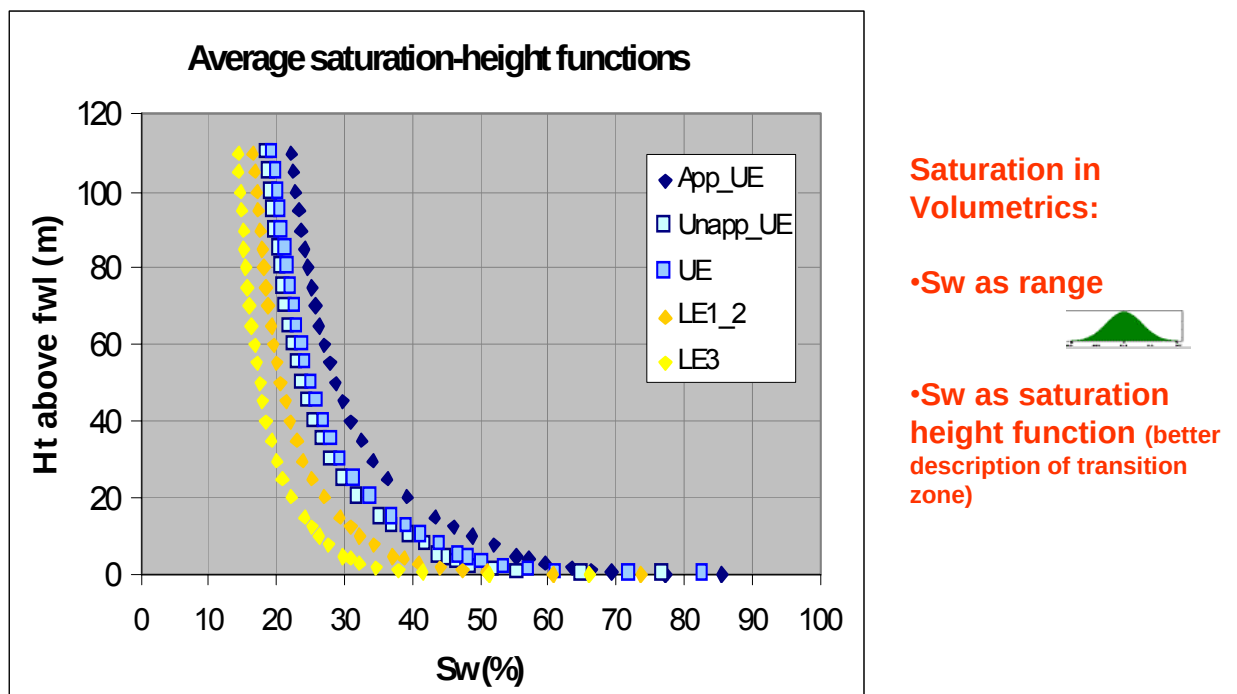
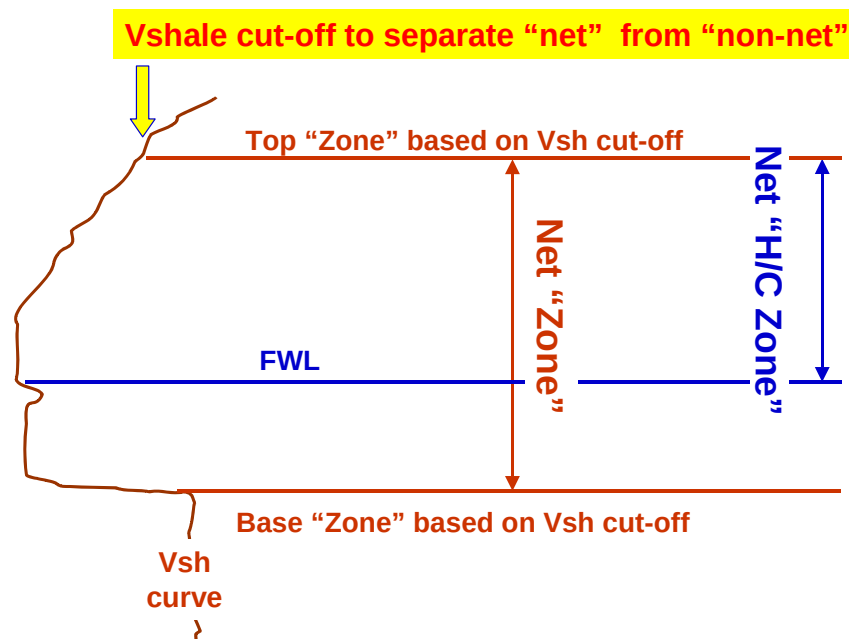


Figure 2: A description of the $(1-S_w)$ input in the STOIIP equation

Best
Practices:
PP Cut-Offs



application of cut-off: only to differentiate between shale and "non" shale

net/gross, Φ , Sw (or SHF)

Figure 3: The selection of the "storage" cut-off

Accuracy: how close to the true value....generally dependent on how well the systematic error is compensated for and controlled

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Precision: how exact....random error....closeness of several measurements made with similar methods -> standard dev.

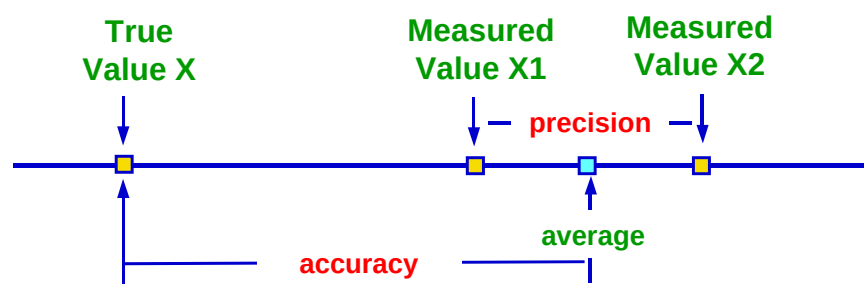


Figure 4: The difference between random and systematic errors